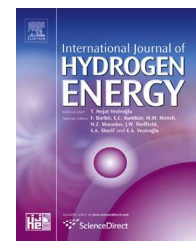


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Feasibility study of methanol production from different renewable sources and thermo-economic analysis

M. Rivarolo*, D. Bellotti, L. Magistri, A.F. Massardo

TPG, University of Genoa, Via Montallegro 1, 16145 Genoa, Italy

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ABSTRACT

This paper aims to present a thermo-economic approach for methanol production comparing different renewable energy sources. In this study, the methanol is produced from the CO₂ hydrogenation in a pressurized reactor; two different plant configurations are analyzed: in the first carbon dioxide is obtained from biogas upgrade, while in the second one CO₂ is acquired from external sources. Carbon dioxide is mixed with hydrogen and then the gas is sent into the reactor for methanol synthesis. Hydrogen is produced by an alkaline pressurized electrolyzer (1 MW, 30 bar) fed by time-dependent electrical energy produced by renewable plants (hydroelectric, wind or photovoltaic), when available; in the remaining periods, electricity is purchased by the national grid.

Since the available electrical energy from the different renewable sources is not constant throughout the year, a time-dependent hierarchical thermo-economic analysis is performed in order to investigate the best plant configuration. The analyses are performed with the W-ECOMP simulation tool (developed by the authors' research group) considering different costs of electricity and different methanol selling prices taking into account the future market forecast. The results for the two plant lay-outs are compared from energetic, economic and environmental point of view for the different renewable energy sources to evaluate the best solution in the Italian scenario.

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Introduction

In the last years, the European Union has promoted the production of innovative fuels (bio-fuels) as one of the main targets for 2020 [1], in order to decrease pollutant emissions from traditional fossil fuels. Methanol is a fuel that presents excellent combustion properties: although it has only about half of the gasoline energy density (20.1 MJ/kg for methanol,

44.3 MJ/kg for gasoline, as reported in Ref. [2]), methanol is a very powerful fuel thanks to its higher octane number [3] (108 for methanol, 95 for gasoline), meaning that the fuel allows for higher compression ratio than traditional gasoline: as a consequence, combustion is more efficient and the emissions in terms of CO and CO₂ are reduced [2–4] compared to traditional fuels (gasoline and diesel). Methanol finds important employment in the chemical industry as well, mainly in the synthesis of formaldehyde, MTBE and acetic acid [5].

* Corresponding author. Tel.: +39 0103532463.

E-mail address: massimo.rivarolo@unige.it (M. Rivarolo).

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Nomenclature

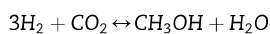
Abbreviation

AEC	alkaline electrolytic cell
CCS	carbon capture sequestration
DPBP	discounted pay back period
NPV	net present value
PEC	purchased equipment cost
RES	renewable energy sources
TCI	total capital investment
TPG	thermochemical power group
W-ECOMP	web-based economic cogeneration modular program

Symbols

H	geodetic height, m
P	power, W
Q	volumetric flow rate, m ³ /s
v	wind velocity, m/s
η	efficiency
ρ	density, kg/m ³

Currently, methanol is mostly produced from fossil fuels: more than 85% of global methanol production is based on natural gas reforming and coal gasification as well; the related synthetic gas is mostly composed by CO and H₂ and the reaction takes place over copper/zinc oxide catalyst. Syngas-based production of methanol is inevitably connected to the emission of a large amount of greenhouse gases (about 3.8 kg CO₂/kg methanol for the coal-based process and about 1.6 kg CO₂/kg methanol for the syngas from natural gas) [6]. Furthermore, the consumption of natural gas and coal (two of the most important energy sources) is significant. The production of methanol from renewable energy sources (RES) allows reducing the intensity of fossil fuels as the main feed of methanol plant, mitigating the CO₂ emission as well. In the case under investigation, the methanol is produced from carbon dioxide and hydrogen obtained by electrolysis and employing renewable sources. The methanol synthesis reaction is reported below [7,8]:



The methanol reaction is exothermic ($\Delta H_r = -49.2$ kJ/mol), meaning that higher methanol yields are obtained at low temperatures and high pressures. Industrially, the ranges of operating temperature and pressure are 200–300 °C and 50–100 bar respectively and the catalyst commonly used is CuO/ZnO/Al₂O₃ based.

A worthy option for renewable methanol generation consists in generating hydrogen by water electrolysis employing electrical energy produced from renewable generators (i.e. wind and hydraulic turbines, photovoltaic panels) [9].

Considering the Italian scenario, the impact of RES has become more and more relevant in the last years: the production from RES has increased from 55 TWh in 2004 to 112 TWh in 2013 (which represents about 38.6% of the national electrical demand) [10]. The main contribution in terms of

production from renewable sources is represented by hydroelectric facilities (52.7 TWh) [10]. Despite the largest part of renewable electricity is generated by hydraulic plants, the impact of other sources has increased significantly in the last years: in particular, wind energy production has increased from 1.8 TWh in 2004 to 14.9 TWh in 2013, while solar energy has increased from a negligible value in 2004 to 21.6 TWh in 2013 [10,11]: at the present day, about 90% of the production from RES is represented by hydroelectric, solar and wind facilities.

In terms of installed power, about 50.2 GW are installed in Italy in 2013 [10,11]: the highest contributions are due to hydroelectric (18.4 GW), solar photovoltaic (18.4 GW) and wind energy (8.6 GW) plants. Considering the number of renewable plants installed in Italy in 2013 [10,11], the very largest contribution is given by photovoltaic (591,029 plants), while the contribution of hydroelectric (3250 plants) and wind energy (1386 plants) is lower.

Taking into account the representative aspects of the Italian scenario reported above, solar, wind and hydraulic energy have been considered as possible renewable electricity inputs for the methanol synthesis system presented in this paper. It is worth remarking that hydroelectric plants are usually affected by water flow availability, which is typically variable on monthly basis; therefore their production is more stable and reliable compared to solar and wind sources, which have hourly/daily variability. From this point of view, methanol can represent a very good solution to store surplus of electrical energy produced by RES, producing methanol that can be used mixed with traditional fuels in order to reduce CO₂ emissions and improve the efficiency [2–4].

In this paper methanol generation from RES is investigated throughout a thermo-economic analysis, assuming different 3 MW renewable energy plants (solar, wind and hydroelectric). The analysis is carried out employing the W-ECOMP software (Web-based Economic Cogeneration Modular Program) for time-dependent thermo-economic optimization: two different plant lay-outs are investigated, considering different economic scenarios and comparing the results for the different RES reported above.

Plant lay-outs

In the present work, two different methanol production plants, employing renewable energy, are investigated, using the W-ECOMP time-dependent thermo-economic optimization approach:

1. Methanol production plant from bio-methane plant;
2. Methanol production plant with pure CO₂ purchasing.

For each plant lay-out, the influence of three different RES (wind, solar and hydroelectric) is taken into proper account, in order to investigate and compare their influence on the process management. The RES have a strong variability, but their distribution is different during the day and year. Wind energy production is determined by the availability of the wind that is completely stochastic and does not have a daily or annually recognizable profile; photovoltaic production depends on

solar energy, it has a characteristic daily profile but its intensity changes during the year; hydroelectric energy production is influenced by water availability throughout the different periods of the year: it can be considered almost constant during the day, but typically variable on a weekly/monthly period. Consequently, the availability of energy depends on the source type and the operative management of the plant changes. The average power distributions throughout the year, based on average monthly values, are reported in Fig. 1 for the three renewable sources.

Methanol production from bio-methane plant

The biogas produced by an anaerobic fermentation system (about 230 Nm³/h) and composed by 52% CH₄ and 48% CO₂ (plus impurities) is sent to a gas cleaning section where the CO₂ (needed for the methanol synthesis) is separated by a CCS system. The hydrogen is produced by a 1 MW pressurized (30 bar) alkaline electrolyzer (AEC) fed by renewable electrical energy (installed power 3 MW): since the renewable sources are intermittent and stochastic, the possibility of purchasing the electricity from the grid, when the renewable energy is not available, is taken into account.

The two reagents, H₂ and CO₂, are sent in the properly volume ratio (3:1) to the mixer and then to the methanol reactor that works at the operating condition of 240 °C and 80 bar [12]. The oxygen co-produced is not employed in the process, but it is assumed to be sold to industrial users. The simplified plant lay-out is shown in Fig. 2.

The plant under investigation consists of the following main components: (i) renewable system; (ii) H₂ production system; (iii) biogas inlet; (iv) gas cleaning section; (v) gas mixer (vi) methanol production system. All these modules have been implemented in W-ECOMP, in order to optimize plant size and management.

The first component simulates the renewable power plant operation: depending on the generator type taken into account, the module represents a wind turbine, a photovoltaic system or a hydroelectric plant. Starting from the energy

availability curve, for each period of the year, the module calculates the electrical energy outlet according with the functions reported below:

- Wind: $P = \frac{1}{2} \rho v^3 \cdot \left(\pi \cdot \frac{D^2}{4} \right) \cdot \eta$
- Photovoltaic: $P = \text{Irradiation} \cdot \text{Area} \cdot \eta$
- Hydroelectric: $P = Q \cdot \rho g H \cdot \eta$

This module presents as output the electrical energy produced. H₂ production module simulates AEC system behavior, using as input the electrical energy produced by the renewable generator or purchased by the national grid. The module gives as output the H₂ production vs. time; it needs as input AEC efficiency and it calculates the power required for the AEC (1 MW size) and for the H₂ compression system. Cleaning section module simulates a Carbon Capture System (CCS) behavior, using as input the biogas flow rate. The module calculates the amount of captured CO₂ in function of the kind of treatment; it presents two outputs: (i) CO₂ flow rate (ii) purge gas flow rate (CH₄). The module needs as inputs the size of the system and the treatment kind option. The sequestered CO₂ is sent to a mixer and then to the chemical reactor: in this model, the stoichiometric ratio between CO₂ and H₂ flow rates (produced by AEC) is set as an input. Penalty costs are applied if the optimal ratio is not achieved. Downstream the mixer, the gas, composed by 75% H₂ and 25% CO₂ (molar fraction), enters the reactor for methanol synthesis. Methanol production module simulates a reactor behavior: the module calculates the amount of CO₂ needed (7.3 times the amount of hydrogen in mass terms) and gives as output produced CH₃OH vs. time.

In this case the biogas production system is not simulated but it is considered as only an input of biogas flow rate.

Methanol production plant from purchased CO₂

This plant is similar to the previous one but in this case the CO₂ is purchased by a plant located nearby or it is captured in

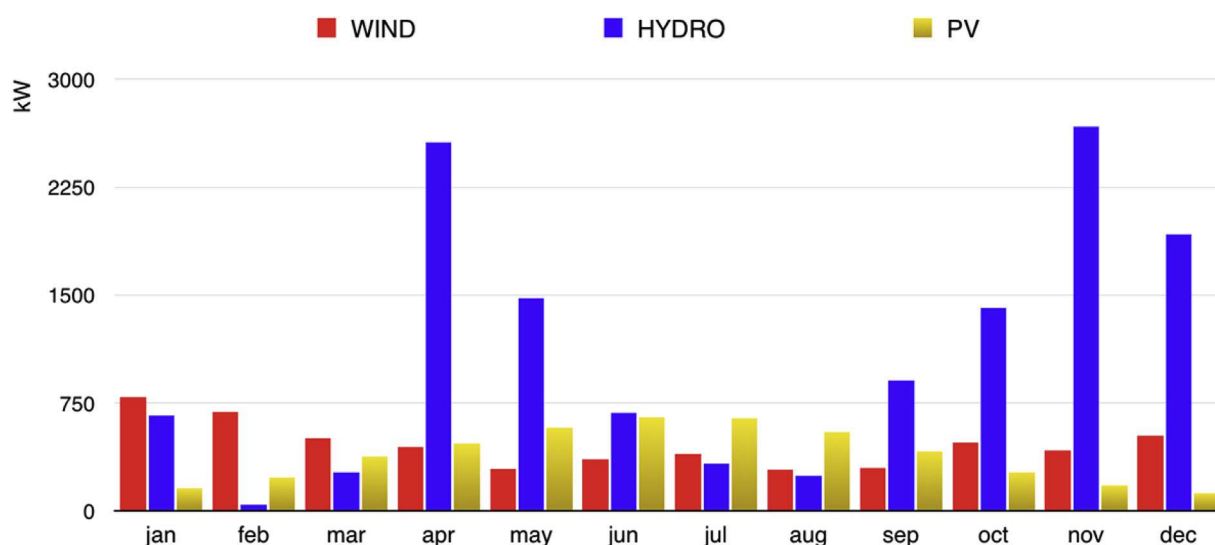


Fig. 1 – Monthly average power distribution of renewable energy sources (wind, photovoltaic, hydroelectric) [10].

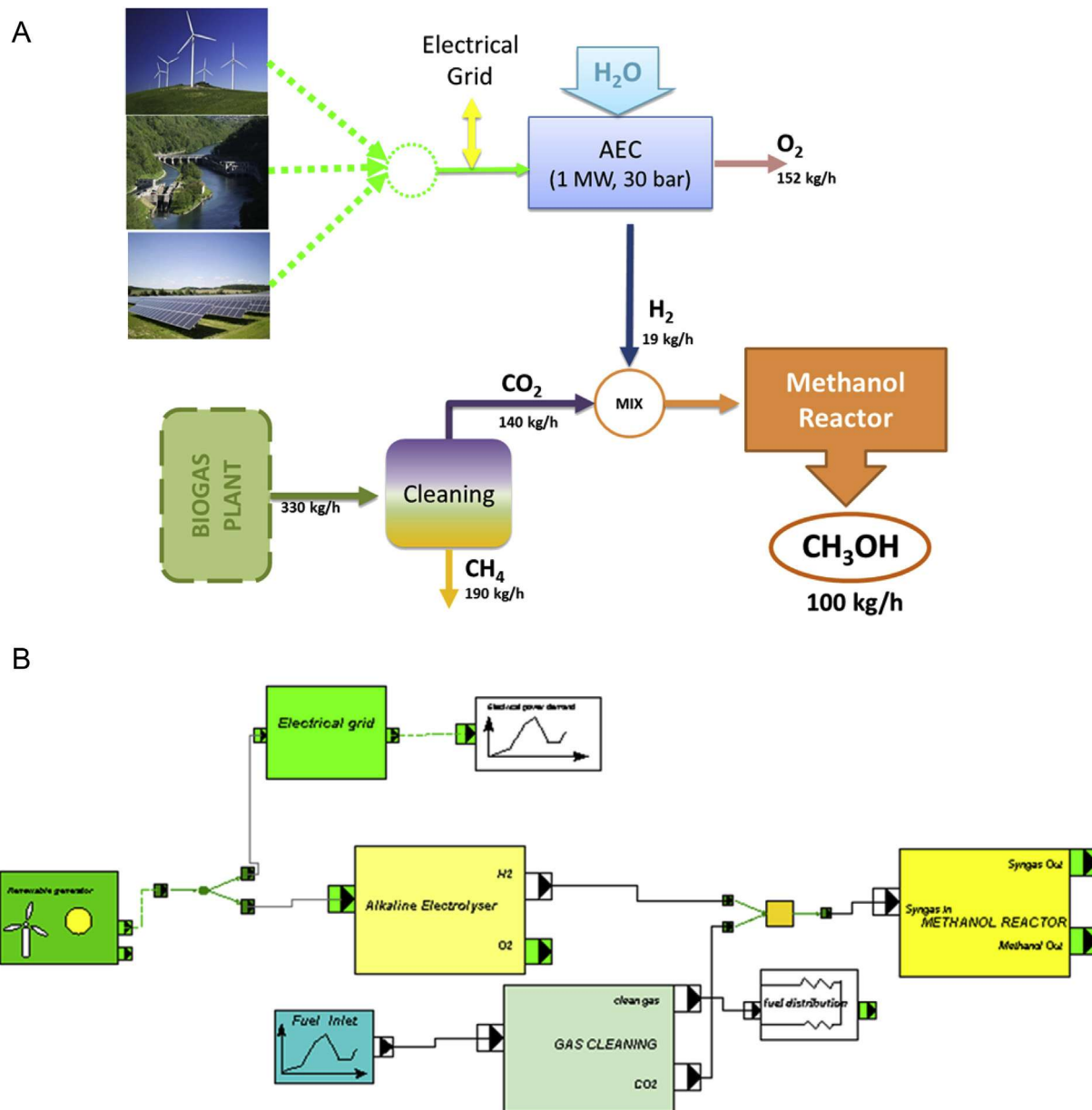


Fig. 2 – Methanol production from biogas (A); plant lay-out in W-ECOMP (B).

another location and then transported in tanks to the methanol synthesis plant. The simplified plant lay-out is shown in Fig. 3.

The plant under investigation consists of the following main modules: (i) renewable generator; (ii) H_2 production system; (iii) CO_2 inlet; (iv) gas mixer (v) methanol production system. All these modules have been implemented in W-ECOMP, as described in the previous paragraph, in order to optimize plant size and management.

Renewable generators, AEC and methanol production modules are the same of the previous configuration. The gas cleaning module is not included anymore; therefore, the plant is simpler and the Total Capital Investment (TCI) is lower; on the other hand, in this case only methanol and oxygen can be sold to the users, while in the biogas configuration methane is

a valuable product as well. The mixer for chemical reactor is the same of the previous case: the carbon dioxide and the hydrogen (produced by AEC at 30 bar) are mixed in a stoichiometric ratio. Penalty costs are applied if the optimal ratio is not achieved. At last the mixture is sent to the methanol production module.

Plant analysis is performed on one year basis, taking into account the time dependent nature of renewable electricity available to the AEC. The aim of the study is the thermo-economic optimization of the plant in order to determine the best management strategy for each configuration and compare the different RES, minimizing annual costs. In addition, the influence of oxygen selling on the results is investigated for each solution.

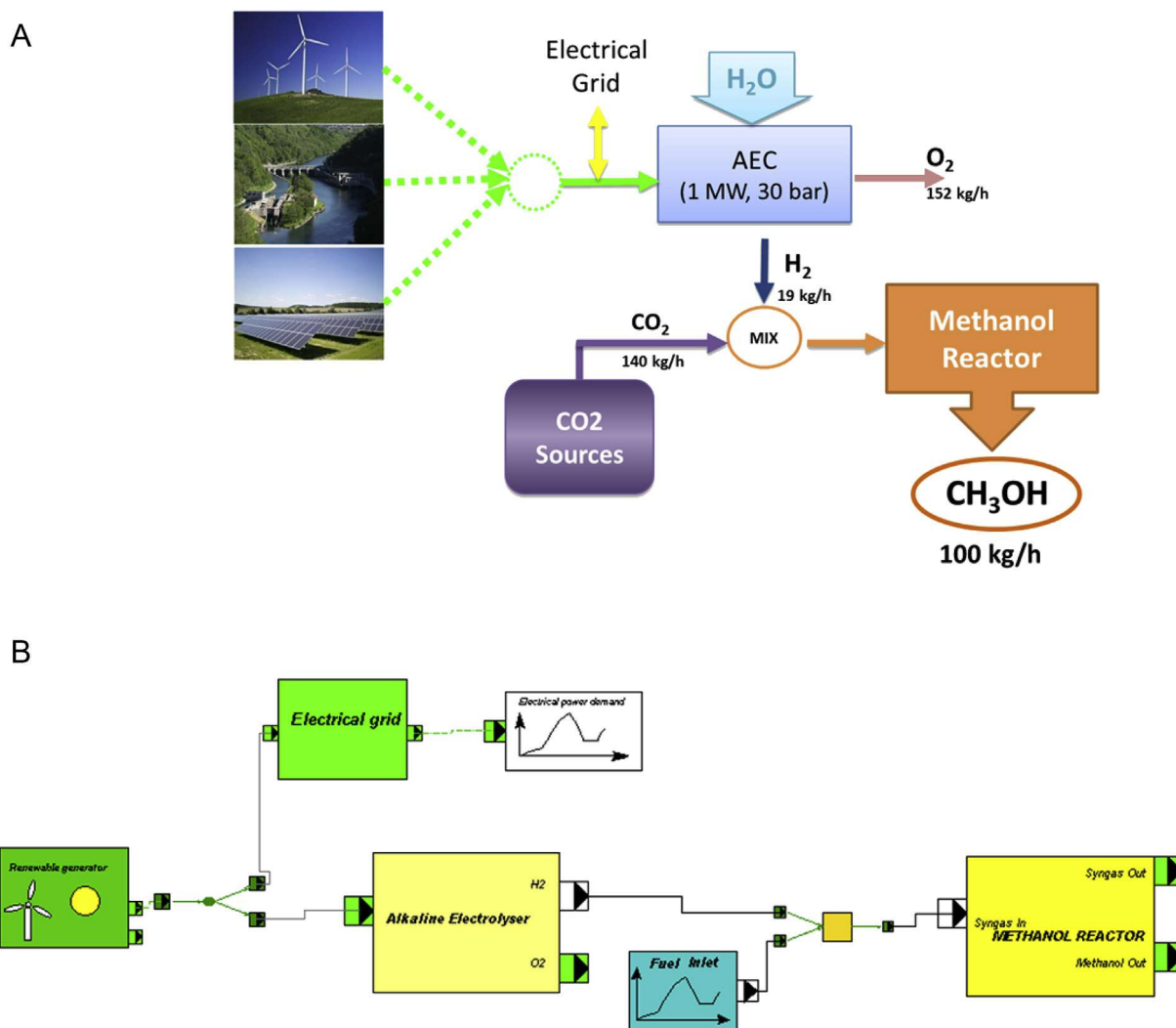


Fig. 3 – Methanol production from pure CO₂ (A); plant lay-out in W-ECOMP (B).

Economic assumptions

To perform the analysis, the Italian economic scenario is chosen and the following economic assumptions are considered:

- **Electricity cost** represents a term of primary importance to find out optimal system configuration. Three different electricity costs of 0.03–0.04 and 0.05 €/kWh have been considered [12].
- **Methanol selling price** is another term that affects significantly the thermo-economic analysis; three different methanol selling prices have been considered, in order to investigate the possibility of an increase in the methanol selling price in response to an increase in the market demand. The methanol selling price considered are 400–500 and 600 €/ton respectively [13,14].
- **Bio-methane selling price** is assumed to be 0.7 €/kg, which is a realistic value based on the selling price to the national gas-grid [15].

- **Oxygen selling price** is assumed 150 €/ton, which represents the minimum selling price for industrial use of oxygen (rates are higher for medical use) [16].
- **CO₂ cost** is affected by transportation technology, as reported in Refs. [17,18]: a cost of 10 €/ton is assumed.

W-ECOMP thermo-economic optimization approach

Considering the time-dependent nature of the problem under analysis, the electrical energy feeding electrolyzers is produced by stochastic and not controllable renewable sources (hydraulic, wind, solar energy), a thermo-economic time-dependent analysis results necessary to investigate and compare the different solutions: the simulations are performed using the W-ECOMP software (Web-based Economic Cogeneration Modular Program); it has been developed by Thermo-chemical Power Group (TPG) at University of Genoa

for thermo-economic analysis of poly-generation power plants under time-dependent conditions [19].

W-ECOMP has a modular approach and a standard component interface, which allows the user for building up and analyzing a wide range of different kinds of energy systems. This approach maintains the flexibility and the extendibility of library components allowing users to add new components without modifying the core of the program. Each component is described by three subroutines, which define mass and energy flows, performance curves in off-design conditions, variable and capital costs [19–21].

The capital cost of each component of plant is evaluated thanks to the cost equations implemented on W-ECOMP software: cost functions are updated periodically (once a year) to take into proper account market prices variations. The most of the cost functions are determined thanks to private communications from TPG's industrial partners and by referring to literature data [21–24]. Table 1 reports the cost functions implemented in the software for the main components of methanol plant.

Cost functions are mandatory to calculate the Purchased Equipment Cost (PEC) for each component; PEC represents the first step to calculate the Total Capital Investment (TCI): TCI is defined by adding to the PEC the terms reported in Table 2, with the same methodology adopted in previous thermo-economic analyses of this kind of energy systems [16,24]. The values below represent the input economic data in W-ECOMP and they are related to the Italian economic scenario considered [24,25] and to the kind of plant analyzed in this study.

The final aim of the economic analysis is the determination of the investment's profitability, taking into proper account the initial investment and the associated risks that are strictly related to the economic scenario and to the kind of plant under analysis. The profitability of the plant is described mostly by two economic parameters: Net Present Value (NPV) and Discounted Pay Back Period (DPBP). NPV is defined as the difference between the sum of all discounted net cash flows and the initial Total Capital Investment (TCI), calculated on the whole plant life.

DPBP is the period of time required to recover the initial investment through the cash inflow given by the plant (considering money discount). Both NPV and DPBP are used to compare different project solutions [26].

In order to find the best solution, W-ECOMP software is provided of a hierarchical strategy with two different optimization levels: the low level is employed for already existing plants, the components size is fixed and the best operational management is determined by a specific genetic algorithm [21,24]; the high level is employed for the design of new plants, defining the optimal size of the different components. At the

both optimization levels, the software performs the minimization of an objective function, taking into proper account variable and fixed annual costs of the plant's components. The description of the two different optimization levels and of the objective function is illustrated in more detail in previous publications of the authors [21,24,27].

Thermo-economic results

Two different plant lay-outs are investigated in the present work, aimed at the production of methanol employing H_2 from AEC (feed by renewable energy) and CO_2 obtained by biogas upgrade into bio-methane (CO_2 is separated in the process) or purchased by an external source respectively. For both plant configurations different energy and economic options are investigated.

Since renewable electricity is a strong time-dependent parameter, affected by the source (wind, solar, hydro) availability, the alkaline electrolyzer is fed by renewable electricity, when available; in the remaining periods the AEC is active by purchasing electricity from the national grid, in order to maximize the production of methanol. Moreover, the high amount of oxygen co-produced by water electrolysis process (8 kg for each kg of H_2 , at 30 bar as well), is sold to different possible users (i.e. steel industries, hospitals). Plant analysis is carried out on a complete year time period, taking into proper account the time dependent nature of the problem, related to variation of renewable electricity available to the AEC.

As far as the electricity purchasing by the national grid is concerned, three different prices are considered (30–40–50 €/MWh), in order to investigate the influence of the electricity price on the economic results [12].

As far as methanol market price in Europe is concerned, the actual average price is 400 €/ton [13]. However, it is worth remarking that methanol demand is getting higher and higher [14]; in particular, methanol utilization as fuel in methanol-gasoline blends may lead to higher methanol selling prices in a low to midterm economic scenario. For this reason, higher methanol selling prices (500 and 600 €/ton) are taken into account, in order to investigate possible next future scenarios.

In order to facilitate the results discussion, the investigated scenarios, described in section 2, are identified as reported in Table 3: **scenario 1** is related to methanol production system employing the CO_2 obtained by biogas upgrading; **scenario 2** is related to methanol plant where the necessary amount of CO_2 is purchased from an external producer.

The results, obtained for both the configurations, are presented below.

Fig. 4 shows the difference between the two plant configurations in terms of TCI: in particular, the solution with methanol production from biogas presents an higher TCI (+50%) than the configuration with CO_2 purchasing. The AEC and the methanol reactor capital costs are the same, the strong difference is represented by the gas cleaning section. For biogas, the upgrade to bio-methane and the separation of CO_2 is quite a simple process and the technologies are already known and used for small size (<1 MW) biogas plants [10], however the capital costs are not negligible. On the other hand, it is worth noting that, in the biogas configuration, the

Table 1 – Cost functions for methanol production plants [20,21,24].

Plant component	Cost function
Pressurized alkaline electrolyzer	$C = 37455 \cdot P^{0.5168}$
Compressors and methanol synthesis	$C = 14.2 \cdot 10^6 \cdot \frac{M_{in}^{0.65}}{54000^{0.65}}$
CO_2 separation/biogas upgrading	$C = 73226 \cdot M_{in}^{0.4859}$

Table 2 – Economic data in W-ECOMP.

Parameters for TCI evaluation			
Startup costs	5% of FCI	Economic data reference year	2014
Working capital	5% of FCI	Construction starting year	2014
Engineering and supervision	10% of PEC	Construction time	1 year
Construction costs	1% of DC	Plant lifetime	15 years
Civil and architectural work	1% of PEC	Depreciation time	10 years
Service facilities	1% of PEC	Inflation rate	0.4%
Purchased equipment cost installation	2% of PEC	Nominal escalation rates	2.5%
Piping	5% of PEC	O&M factor	1.04
Instrumentation and controls	6% of PEC	Average income tax rate	22%
Electrical equipment and materials	6% of PEC		
Contingencies	10% of CC + DC		

separation process allows for obtaining bio-methane (CH_4 content higher than 95%), which represents a valuable product that can be sold to industrial/civil users in order to increase the profits.

As far as electrical production is concerned, it is worth noting that the equivalent operative hours (defined as the ratio of annual electrical energy production and installed power) are not the same for the different considered RES: the amounts of equivalent hours are about 3200 h for hydroelectric, 1300 for wind and 1100 for photovoltaic. Referring to the contribution of renewable electricity to the total annual input required by the AEC and the compressors, hydroelectric energy assures about 67% of the total electrical input required by the system, meaning that the most of methanol is produced by renewable sources; in the case of wind and solar energies, since the availability of the sources are lower, their contributions are lower as well (30% in case of wind, 27% in case of solar).

Fig. 5 reports the contribution of renewable electrical energy for the three different RES in typical winter (December) and summer (June) days: the advantage of employing hydroelectric energy is evident. Since hydraulic source is more predictable, it allows for a constant contribution of RES in the different hours of the day; on the other hand, wind and solar energy are highly stochastic, meaning that, in case of scenarios where electrical energy purchasing is not allowed (i.e. systems that are not connected to the national grid) or not convenient (high cost of electrical energy), the devices would work with frequent on-off that can reduce their lifetime (in case of AEC) or cause management problems and thermal stress (in case of methanol synthesis reactor).

Fig. 6 shows the economic results, in terms of DPBP, related to the methanol production from the different RES, for both the configurations; the RES production is integrated with the electrical energy purchased from the grid in order to always keep the AEC at nominal conditions. The influence of electricity cost and the kind of RES considered are investigated: only the solutions with DPBP lower than 8 years (which represents about the half of plant's lifetime) are considered acceptable. At the current selling price of CH_3OH , the hydroelectric source only is economically feasible; photovoltaic and wind are not convenient, considering that their low number of operating hours leads to higher costs for electricity purchasing. Fig. 6 shows the superiority of obtaining CO_2 from biogas upgrading (scenario 1), compared to the CO_2 purchasing solution (scenario 2).

Fig. 7 reports the comparison between costs and revenues for both the plant configurations, considering the scenario with low prices for electricity (30 €/MWh) and methanol (400 €/ton). As far as the costs are concerned, the electrical energy purchasing amount is the same in both the cases; on the other hand the annual fixed cost results higher in the biogas configuration case, for the reasons explained above (see Fig. 4). As far as the revenues are concerned, it is worth noting that, in the biogas configuration, the main income is represented by the methanol selling but the bio-methane is a significant part of the revenues as well. The oxygen selling may represent an important additional income, depending on the market. In the CO_2 purchasing configuration, the whole revenues are represented by the methanol and the oxygen selling. The revenues related to the biogas configuration result higher but the costs are higher as well, due to the higher TCI: globally, the methanol production from biogas plant (scenario 1) represents the more feasible solution from the economic point of view, as reported in Fig. 6 as well: however, the biogas solution presents higher investment cost and complexity of the plant, compared to the CO_2 purchasing solution (scenario 2).

Future scenario for methanol

At last, the influence of methanol selling price on the economic results is investigated. Since methanol market is growing significantly and methanol demand is getting higher and higher worldwide, the selling price will probably increase as well. In order to investigate the influence of this fact, a further analysis is performed considering methanol selling prices of 500 and 600 €/ton respectively.

Fig. 8 shows the results of the analysis in terms of DPBP for the two different scenarios, considering an electricity cost of 30 €/MWh: in a low to midterm future scenario, increasing CH_3OH selling price, DPBP can decrease significantly. It is worth noting that wind and solar renewable sources, which are not convenient at the current methanol market prices, may become economically attractive as well. Comparing the two plant configurations, it is worth observing that, for the highest selling prices considered here, DPBP values for scenarios 1 (CO_2 from biogas upgrading) and 2 (CO_2 purchasing) are nearly the same.

Fig. 9 reports the comparison between TCI and NPV for the two plant lay-outs and for the different methanol prices,

Table 3 – Investigated scenarios for methanol synthesis from renewable sources.

Scenario	CO ₂ Source	RES	Energy option	Electricity purchasing price	Methanol selling price	Oxygen selling price	Bio-methane selling price
1	Biogas plant	Wind	Renewable energy when available; grid	0.03–0.04–0.05 €/kWh	400–500–600 €/ton	150 €/ton	700 €/ton
2	CO ₂ purchasing	Photovoltaic Hydroelectric	energy in the remaining periods	0.03–0.04–0.05 €/kWh	400–500–600 €/ton	150 €/ton	–

considering the hydroelectric source, since it is the most attractive solution. TCI (which is not affected by methanol selling price) is lower for scenario 2, where capital costs for the CO₂ separation system and for biogas upgrading are not included; on the other hand, scenario 1 allows for higher annual revenues due to bio-methane selling and, as consequence, higher NPV are obtained. Comparing the two plant configurations, it is worth observing that for the current methanol price (400 €/ton) scenario 1 allows for the higher NPV/TCI ratio compared to scenario 2, showing that the investment is more profitable; on the other hand, for the highest methanol selling price (600 €/ton), the two solutions are almost equivalent and the NPV/TCI ratio is 2 for both. The different trends of NPV for the two scenarios are related to the kind of products: as Fig. 7 shows, in scenario 2 the most of the revenues are due to methanol sale, therefore its influence is stronger; in scenario 1, the presence of other co-products (bio-methane, oxygen) leads to a lower influence of methanol selling price. In a scenario with high methanol demand, solution 1 may become even more attractive from the economic point of view, considering also the lower investment required.

Environmental analysis

Traditionally, most of the methanol is synthesized from fossil fuels by CH₄ reforming: in the process, high amounts of carbon dioxide are co-produced and high amounts of fossil fuels (methane) are consumed. Considering producing methanol from renewable sources throughout the process described in section 2, two advantages can be obtained from the environmental point of view:

- no fossil fuels consumption (327 ton/year of CH₄ are avoided), since hydrogen is produced in the AEC fed by RES;
- co-production of high amounts of oxygen (1313 ton/year) from the AEC.

Table 4 reports the environmental analysis for the two plant configurations: carbon dioxide emissions related to methanol synthesis are avoided in both the scenarios. However, if CO₂ is purchased by an external source, CO₂ emissions related to methanol combustion (i.e. in case of use in ICEs in blends with gasoline or diesel [1,2]) have to be taken into proper account in the final emissions balance. On the contrary, if CO₂ is obtained by upgrade of biogas produced by anaerobic digestion of agricultural biomass, then the CO₂ impact of the whole cycle is equal to zero. In fact, the amount of carbon dioxide emitted in the combustion is the same sequestered by the biomass during their life cycle and the global balance is zero. Considering that the current carbon tax related to CO₂ emissions are equal to 7 €/ton in the Italian scenario [15], about 30,000 € per year can be saved.

Conclusions

In this paper, a thermo-economic feasibility analysis for methanol synthesis from different RES (solar, wind, hydroelectric) has been performed: methanol is synthesized in a

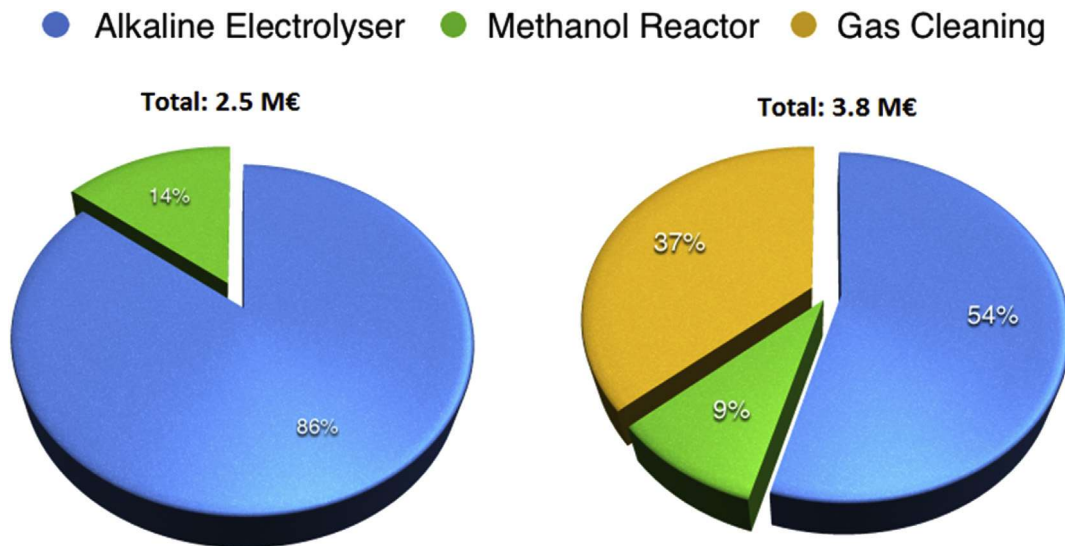


Fig. 4 – Capital costs distribution for methanol production from CO₂ purchasing (left) and biogas (right).

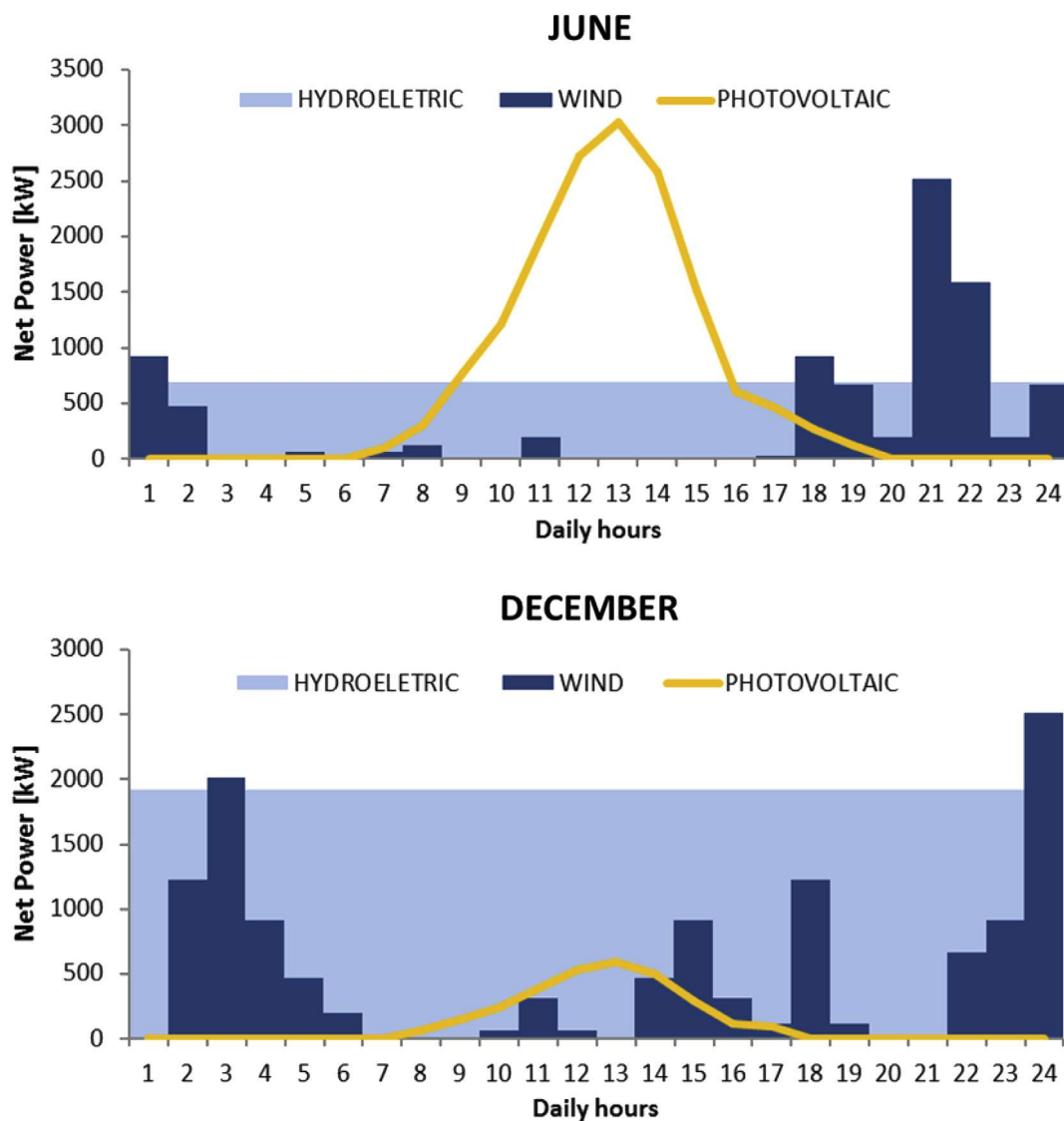


Fig. 5 – Net power average production from different RES in summer (June) and winter (December).

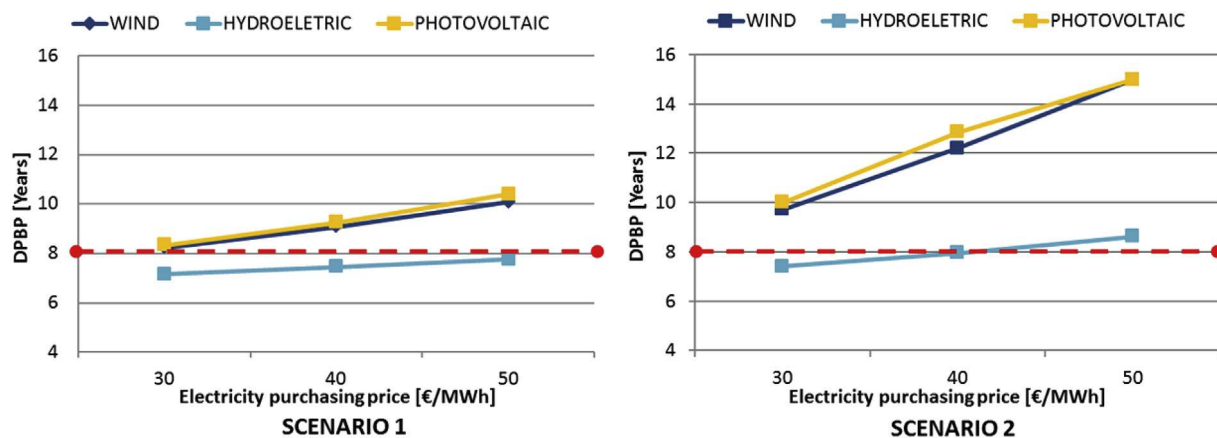


Fig. 6 – DPBP comparison for the different RES and purchasing electricity prices.

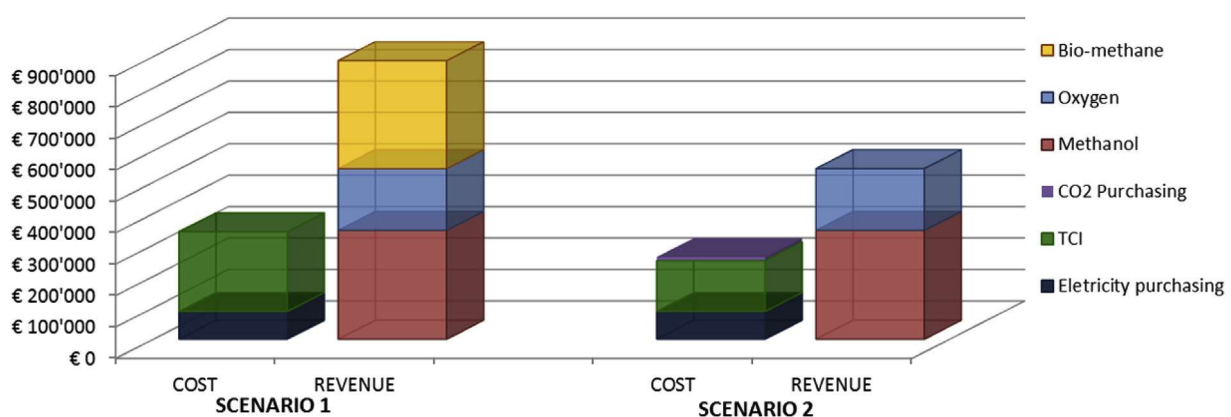


Fig. 7 – Annual costs and revenues comparison for the two scenarios (considering energy purchasing at 30 €/MWh).

chemical reactor, starting from H₂ and CO₂. The hydrogen is produced by a pressurized alkaline electrolyzer (1 MW, 30 bar) fed by electrical energy produced by the different renewable energy plants (3 MW), previously installed, while CO₂ is sequestered from different sources. Two different plant configurations were investigated, namely:

- Integration of H₂ produced by AEC fed by renewable energy with CO₂ separated from biogas produced by an anaerobic digestion process and upgraded to bio-methane;
- Integration of H₂ produced by AEC fed by renewable energy with CO₂ purchased by an external plant.

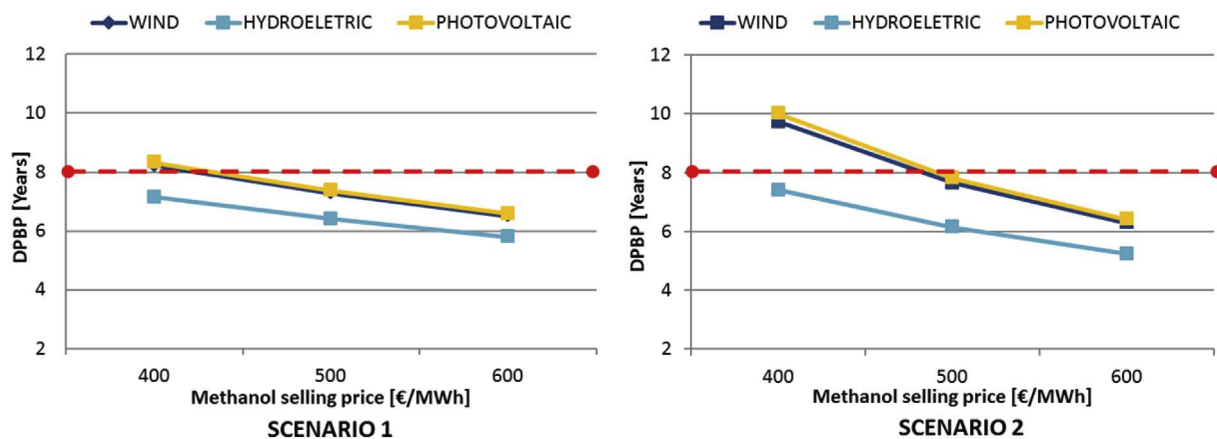


Fig. 8 – DPBP comparison for the different RES and purchasing electricity prices.



Fig. 9 – TCI/NPV comparison for the different methanol selling prices.

The thermo-economic analysis has been performed considering different scenarios, investigating the influence of different electrical energy costs (0.03–0.04–0.05 €/kWh) and selling prices for methanol (400–500–600 €/ton) to simulate next future scenarios. The analysis has been performed using the W-ECOMP software, developed by the Authors research group for thermo-economic optimization of energy systems in time-dependent conditions. The results allow for the following conclusions to be drawn:

- In the Italian scenario the best economic results are obtained with hydroelectric source, which guarantees an higher number of equivalent operative hours (3000) than wind (1300) and solar (1100) energies; when renewable energy is not available, electrical energy to AEC is purchased by the national grid;
- At the current CH₃OH selling price, the plant configuration with CO₂ purchasing gives an acceptable DPBP only for the hydroelectric source and at low electricity purchasing price (30 €/ton); therefore this solution does not seem profitable;
- However, considering future methanol selling prices of 500 €/ton and 600 €/ton [12] the plant configuration has lower DPBP; it is worth noting that coupling the AEC with wind and solar sources becomes economically feasible in these scenarios;
- The plant configuration with CO₂ sequestration by biogas plant is characterized by higher investment costs, due to the plant complexity. On the other hand, revenues are

higher and DPBP is lower than 8 years in the most of the scenarios, thanks to CH₄ selling. At low purchasing electricity cost (0.03 €/kWh), wind and solar are competitive even at the current methanol selling price;

- Despite the biogas configuration gives the best economic performances, it is worth noting that the configuration with CO₂ purchasing allows for a lower TCI and a simpler plant lay-out; furthermore, at higher methanol selling prices, it gives results that are very similar to the biogas configuration;
- From the environmental point of view, both the solutions allow avoiding the consumption of methane and the emissions of CO₂ related to the standard methanol production from fossil fuels; the economic performances can be further improved considering the carbon tax related to CO₂ emissions (about 30,000 € per year).

Results show a great potential for methanol production from renewable sources: the economic performances can be further improved considering the European financial incentives for biofuel production.

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REFERENCES

- [1] Shengua L, Clemente ERC, Tiegang H, Yanjv W. Study of spark ignition engine fueled with methanol/gasoline fuel blends. *Appl Therm Eng* 2007;27:1904–10.
- [2] Balki MK, Sayin C. The effect of compression ratio on the performance, emissions and combustion of an SI (spark ignition) engine fueled with pure ethanol, methanol and unleaded gasoline. *Energy* 2014;71:194–201.
- [3] Celik MB, Ozdalyan B, Alkan F. The use of pure methanol as fuel at high compression ratio in a single cylinder gasoline engine. *Fuel* 2011;90:1591–8.

Table 4 – Environmental analysis.

		Methanol from fossil fuels	Scenario 1	Scenario 2
CH ₄ consumption	ton/year	327	–	–
O ₂ production	ton/year	–	1313	1313
CO ₂ emission (producing CH ₃ OH)	ton/year	1855	–	–
CO ₂ emission (employing CH ₃ OH)	ton/year	2460	–	2460
CO ₂ total emission	ton/year	4315	–	2460

- [4] Gong CM, Huag K, Jia JL, Su Y, Gao Q, Liu XJ. Improvement of fuel economy of a direct-injection spark-ignition methanol engine under light loads. *Fuel* 2013;90:1826–32.
- [5] Pellegrini L, Soave G, Gamba S, Langè S. Economic analysis of a combined energy–methanol production plant. *Appl Energy* 2011;88:4891–7.
- [6] Galindo Cifre P, Badr O. Renewable hydrogen utilisation for production of methanol. *Energy Convers Manag* 2007;48:519–27.
- [7] Van-Dal ES, Bouallou C. Design and simulation of a methanol production plant from CO₂ hydrogenation. *J Clean Prod* 2013;57:38–45.
- [8] Boretti A. Renewable hydrogen to recycle CO₂ to methanol. *Int J Hydrogen Energy* 2013;38:1806–12.
- [9] Sayah AK, Sayah AK. Wind-hydrogen utilization for methanol production: an economy assessment in Iran. *Renew Sustain Energy Rev* 2011;15:3570–4.
- [10] http://www.terna.it/default/Home/SISTEMA_ELETTRICO/statistiche/dati_statistici.aspx [accessed 30.05.15].
- [11] GSE (Gestore Servizi Energetici). Rapporto Statistico 2013 Impianti a fonti rinnovabili. <http://www.gse.it/home> [accessed 01.06.15].
- [12] <http://www.mercatoelettrico.org/it/> [accessed 30.05.15].
- [13] <http://www.methanex.com/products/methanolprice.html> [accessed 30.05.15].
- [14] <http://igpenergy.com/products/methanol-overview/methanol/> [accessed 30.05.15].
- [15] Private communication by Italian Energy National Authority (Autorità per l'Energia Elettrica, il Gas e il Sistema Idrico, AEEGSI, <http://www.autorita.energia.it>).
- [16] Rivarolo M, Bellotti D, Mendieta A, Massardo AF. Hydro-methane and methanol combined production from hydroelectricity and biomass: thermo-economic analysis in Paraguay. *Energy Convers Manag* 2014;79:74–84.
- [17] PlantsKlokka Ø, Schreiner PF, Pagès-Bernaus A, Tomasgard A. Optimizing a CO₂ value chain for the Norwegian Continental Shelf. *Energy Policy* 2010;38:6604–14.
- [18] Gao L, Fang M, Li H, Hetland J. Cost analysis of CO₂ transportation: case study in China. *Energy Procedia* 2011;4:5974–81.
- [19] www.tpg.unige.it/w-ecomp.php [accessed 30.06.15].
- [20] Rivarolo M, Bogarin J, Magistri L, Massardo AF. Time-dependent optimization of a large hydrogen generation plant using “spilled” water at Itaipu 14 GW hydraulic plant. *Int J Hydrogen Energy* 2012;37:5434–43.
- [21] Rivarolo M, Massardo AF. Optimization of large scale bio-methane generation integrating “spilled” hydraulic energy and pressurized oxygen blown biomass gasification. *Int J Hydrogen Energy* 2013;38:4986–96.
- [22] Jensen JO, Bandur V, Bjerrum NJ, Jensen SO, Ebbesen S, Mogensen M, et al. Pre investigation of water electrolysis. *PSO-F&U* 2006-1-6287. 2006.
- [23] Dirkse EHM. Biogas upgrading using the DMT TS-PWS technology. *DMT Environmental Technology*. 2008.
- [24] Bellotti D, Rivarolo M, Magistri L, Massardo AF. Thermo-economic comparison of hydrogen and hydro-methane produced from hydroelectric energy for land transportation. *Int J Hydrogen Energy* 2015;40:2433–44.
- [25] www.tradingeconomics.com [accessed 31.08.15].
- [26] Bejan A, Tsatsaronis G, Moran M. Thermal design and optimisation. Interscience Publication, John Wiley & Sons Inc; 1996.
- [27] Rivarolo M, Greco A, Massardo AF. The thermo-economic optimization of the impact of renewable generators on poly-generation smart grids including hot thermal storage. *Energy Convers Manag* 2013;65:75–83.